



Three Rivers Academy

Cambridge International AS & A Level

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CHEMISTRY

9701/21

Paper 2 AS Level Structured Questions

June 2026

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [].
- The Periodic Table is printed in the question paper.
- Important values, constants and standards are printed in the question paper.

This document has **14** pages. Any blank pages are indicated.

- 1 (a) Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

..... [1]

(ii) Explain how graphite conducts electricity.

.....
..... [1]

(iii) Explain why diamond does **not** conduct electricity.

.....
..... [1]

- (b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

(i) Write an equation for each reaction.

.....
..... [2]

(ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride

silicon(IV) chloride [2]

(iii) Compare the appearance of the mixtures produced when each reaction is complete.

.....
..... [2]

- (c) Some oxides are amphoteric.

(i) Describe what is meant by amphoteric.

.....
..... [1]

(ii) Identify the formula of a Period 3 oxide that is amphoteric.

..... [1]

- (d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point/°C	force of attraction broken during melting
SO ₂	-73	
H ₂ O	0	
SiO ₂	1610	
Na ₂ O	1132	
MgO	2852	

- (i) Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

- (ii) A student suggests the following hypothesis.

*The stronger the covalent bond between atoms in non-metal oxides,
the higher the melting point.*

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

 [2]

[Total: 16]

- 2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(a) (i) Define relative atomic mass.

.....

 [2]

(ii) Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

Table 2.1

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample. Show your working.

relative isotopic mass = [2]

(b) Deduce the number of pairs of electrons in the 3d sub-shell in an iron(II) ion.

..... [1]

(c) Sketch the shape of the lowest energy orbital in the shell with principal quantum number $n = 2$.

[1]

- (d) Complete Table 2.2 to show information about particles in one ion of $^{56}\text{Fe}^{3+}$.

Table 2.2

particle	number of particles present in one ion of $^{56}\text{Fe}^{3+}$
electrons	
	30

[2]

- (e) The atomic radius of iron is $1.26 \times 10^{-10}\text{m}$.

Suggest the change to the radius, if any, after an iron atom reacts to produce an Fe^{3+} ion. Explain your answer.

.....

.....

.....

..... [2]

[Total: 10]

3 (a) C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....
 [1]

(ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.1

[4]

(iii) Explain why the major product of this reaction is 2-bromopropane rather than 1-bromopropane.

.....

 [2]

- (b) $\text{H}_2(\text{g})$ and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

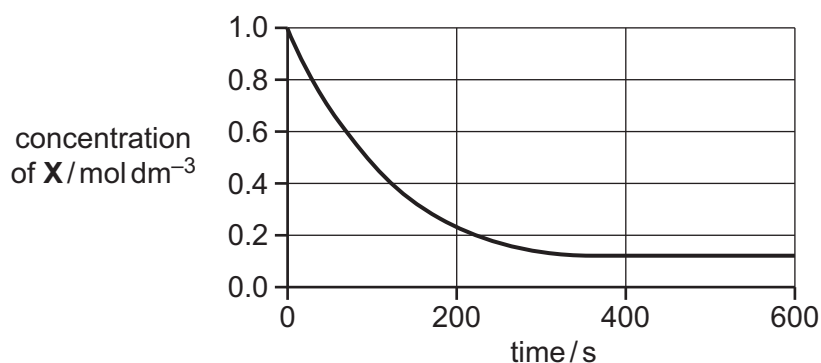


Fig. 3.2

- (i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer.

.....
 [1]

- (ii) Use Fig. 3.2 to describe how the gradient changes as the reaction proceeds. State what this shows about the rate during this reaction.

.....

 [1]

(c) Hydrocarbon **Y** contains two C=C groups. There are no other functional groups present.

Y reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .

Table 3.1

bond	energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
H–H	436
C–H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = kJ mol^{-1} [2]

- (d) Hydrocarbon **Y** reacts with $\text{H}_2(\text{g})$.

Fig. 3.3 shows the distribution of energies of $\text{H}_2(\text{g})$ at temperature T .

Area **A** represents the number of molecules with energy greater than or equal to the activation energy, E_{A} , at temperature T .

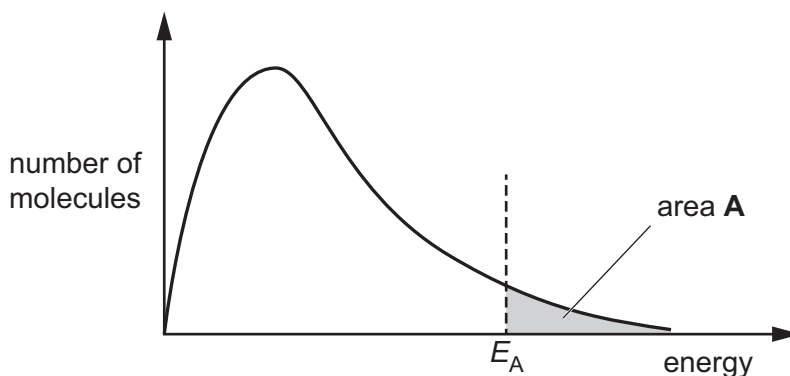


Fig. 3.3

- (i) Annotate Fig. 3.3 to show the effect of adding nickel to **Y** and $\text{H}_2(\text{g})$ at temperature T . [1]
- (ii) Area **B** (not labelled on Fig. 3.3) represents the number of molecules with energy greater than or equal to the activation energy when nickel is added at temperature T .

State the difference, if any, between areas **A** and **B**. Explain the significance of your answer on the rate of hydrogenation of **Y**. Give your answer in terms of collisions.

.....

.....

..... [2]

- (e) Alkene **Z** contains two $\text{C}=\text{C}$ bonds. **Z** reacts with an excess of hot concentrated acidified KMnO_4 to produce **only** CH_3COCH_3 , $\text{HOOCCH}_2\text{COOH}$, CO_2 and H_2O .

Suggest the structure of **Z**.

[2]

[Total: 16]

- 4 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(a) Table 1.1 shows some information about the elements Mg to Ba.

Table 1.1

element	proton number	shorthand electronic configuration	sum of first and second ionisation energies / kJ mol^{-1}	third ionisation energy / kJ mol^{-1}
Mg	12	[Ne] $3s^2$	2186	7740
Ca	20	[Ar] $4s^2$	1740	4940
Sr	38	[Kr] $5s^2$	1608	4120
Ba	56	[Xe] $6s^2$	1468	3390

- (i) Give the full electronic configuration of a calcium atom.

..... [1]

- (ii) Explain the trend shown in the sum of the first and second ionisation energies, as proton number increases.

.....

 [2]

- (iii) Write an equation to represent the third ionisation energy of barium.

..... [1]

- (iv) Explain why the third ionisation energy of magnesium is much greater than the sum of its first and second ionisation energies.

.....

 [2]

(b) When strontium reacts with water, an alkaline solution is produced.

(i) Write an equation for this reaction. Include state symbols.

..... [2]

(ii) Samples of the alkaline solution produced in **1(b)(i)** are tested with dilute HCl(aq) and separately with dilute $\text{H}_2\text{SO}_4\text{(aq)}$.

State what you would observe in each test.

observation with dilute HCl(aq)

.....

observation with dilute $\text{H}_2\text{SO}_4\text{(aq)}$

..... [2]

(c) The nitrates and carbonates of Group 2 elements decompose on heating.

(i) State the trend in the thermal stability of the Group 2 carbonates down the group.

..... [1]

(ii) Write an equation for the thermal decomposition of $\text{Ca(NO}_3)_2$.

..... [1]

(d) Calcium chlorate(I), Ca(ClO)_2 , is used for water purification in the same way as sodium chlorate(I), NaClO .

Ca(ClO)_2 can be prepared by the reaction of aqueous Ca(OH)_2 with Cl_2 .

(i) Write an ionic equation for the reaction of cold dilute OH^- ions with Cl_2 .

..... [1]

(ii) State why NaClO and Ca(ClO)_2 are used in water purification.

..... [1]

[Total: 14]

- 5 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

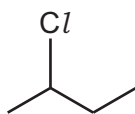


Figure 3.1

- (a) Explain why 2-chlorobutane is classified as a secondary halogenoalkane.

.....
 [1]

- (b) 2-chlorobutane can be formed by the electrophilic addition of HCl to different isomers of C_4H_8 . Compound **A** is one of these isomers.

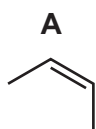
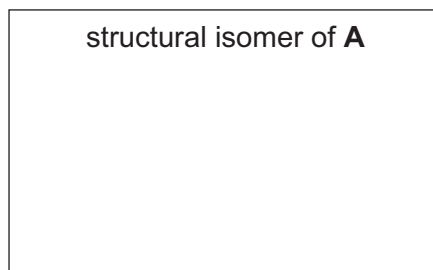


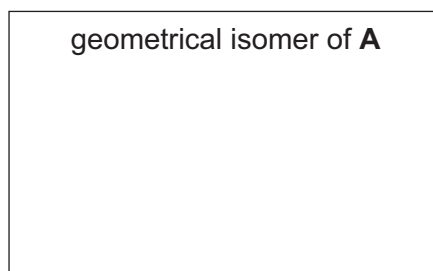
Figure 3.2

- (i) Draw a structural isomer of **A** that can also form 2-chlorobutane when it reacts with HCl .



[1]

- (ii) Draw and give the systematic name of the geometrical isomer of **A**.



systematic name [2]

[Total: 4]

Important values, constants and standards

molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Faraday constant	$F = 9.65 \times 10^4 \text{ C mol}^{-1}$
Avogadro constant	$L = 6.022 \times 10^{23} \text{ mol}^{-1}$
electronic charge	$e = -1.60 \times 10^{-19} \text{ C}$
molar volume of gas	$V_m = 22.4 \text{ dm}^3 \text{ mol}^{-1}$ at s.t.p. (101 kPa and 273 K) $V_m = 24.0 \text{ dm}^3 \text{ mol}^{-1}$ at room conditions
ionic product of water	$K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ (at 298 K (25 °C))
specific heat capacity of water	$c = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$ ($4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

The Periodic Table of Elements

Group																					
1	2	1														13	14	15	16	17	18
		1 H hydrogen 1.0																			